FBLA: MANAGEMENT INFORMATION SYSTEMS

Competency: Systems Analysis & Design (Systems Development Life Cycle)
Tasks
1. Demonstrate knowledge of the key functions and subsystems of the network system.
2. Demonstrate knowledge of the system life-cycle approach and identify and explain the steps in the systems development life cycle.
3. Identify the functions of systems analysts.
4. Select basic fact-gathering techniques to be used and conduct a preliminary investigation.
5. Record facts gathered through the system investigation.
6. Define the scope of the systems project.
7. Identify time, technology, and resource constraints.
8. Perform appropriate diagnostic tests.
9. Investigate system alerts.
10. Design system output, system input, files, and processing.
11. Analyze the interaction of the operating system and hardware architecture.
12. Justify the communications selections for the system (e.g., single PCs, LANS, and/or WANS).
13. Identify the system components and their relationships.
14. Specify the workflow system.
15. Develop programming specifications and program the system.
16. Test and document the system.
17. Design a framework for evaluating information system function and individual applications.
18. Compare the capabilities of an application with the requirements it is intended to meet.
19. Identify alternative outcomes of the application verification process.
20. Evaluate processes and outcomes including the results and probabilities of errors.
21. Modify inputs, outputs, and processing to refine an application.
22. Determine needed follow-up actions including recommendations for new features or enhancements to existing tools.

Competency: Database Management and Modeling Concepts
Tasks
1. Demonstrate knowledge of the features, functions, and architecture of a database management system.
2. Identify the uses of a DBMS in business organizations.
3. Demonstrate knowledge of how a DBMS ensures data integrity through transaction-control techniques.
4. Trace the evolution of DBMS models and their implementation.
5. Produce single- and multiple-level control break reports and subtotal and final totals.
6. Write programs that allow the user to make a menu choice, that require statements to be executed multiple times, and that access multiple files.
7. Design an information system within a database environment.
8. Build database applications and distribute data across a distributed DBMS.
9. Analyze/model organizations using Entity-Relationship and Object technologies.
10. Create/update and query a relational database using Structured Query Language.
11. Manage and monitor implementation of a database management system.
12. Identify and document problems and propose solutions that are congruent with application requirements.
13. Apply databases to actual situations and business problems.
14. Create conceptual data models.
15. Identify and select logical and physical structures appropriate for specific applications.
16. Create and normalize a logical data model in accordance with established company policy.
17. Plan, develop and normalize a database schema.
18. Explain the options for converting legacy records to electronic database management systems.

Competency: Object Oriented Analysis and Design
Tasks
1. Identify and define object-oriented programming terminology.
2. Describe the fundamental object-oriented principles and identify the characteristics and uses of object-oriented processing.
3. Describe the object-oriented software development process.
4. Explain the purpose, activities, and artifacts of the following Object-Oriented Software Development workflows: requirements gathering, requirements analysis, architecture, design, implementation, testing, and deployment.
5. Choose an object-oriented methodology that best suits the project.
6. Create a project vision document from the results of interviews and risk analysis.
7. Document the system in the System Requirements Specifications.
8. Create and refine the diagram for a software system based on the System Requirements Specifications.
9. Identify the key abstractions based on the analysis.
10. Describe the Analysis Model, the Architecture Model and the Component (Design) Model.
11. Construct the problem domain model.
12. Create the Analysis Model using Robustness analysis.
13. Distinguish between architecture and design.
14. Create the Architecture workflow artifacts.
15. Create the architecture model for presentation.
16. Create a solution model for GUI and Web UI application.
17. Refine the attributes, relationships, and methods of the Domain model.
18. Apply design patterns (e.g., composite, strategy, observer, and abstract factory) to the Solution Model.
19. Model complex object state using state chart diagrams.

Competency: User Interfaces
Tasks
1. Define hardware-software interface issues for a system.
2. Describe interface techniques and standards.
3. Demonstrate knowledge of version management and interface control.
4. Assess the impact of changes that affect interfaces.
5. Integrate human factors and user interfaces in visual design.
6. Develop user interfaces.
7. Develop programs that interface with a data store.
8. Understand the characteristics of potential users, their tasks, and their environments.
9. Relate to the ways in which the users define themselves and their roles (e.g., jobs, tasks, and tools they use).
10. Conduct tasks analysis to review the workflow and other aspects of the user's job.
11. Interpret the results of tasks analysis.
12. Select techniques that are appropriate to a project and the user's environment.
13. Analyze and document data by creating representations such as work flows, task hierarchies, and task scenarios.
14. Reorganize results using such techniques as affinity diagrams and insight sheets to clarify relationships.
15. Form the design using storyboarding, sketching, and video presentations.
16. Test and document user interface usability.
Competency: System Controls
Tasks
1. Set up/maintain user accounts on multiple systems.
2. Provide technical product support and facilitate the delivery of technical services.
3. Manage inventory and assets.
4. Participate in evaluation of the total system.
5. Identify new application requirements within the system.
6. Document performance problems.
7. Analyze historical data to identify trends.
8. Formulate technical procedures.

9. Prepare documentation manuals.
10. Prepare required reports.
11. Apply data structure concepts to the storage and retrieval of data (e.g., map a model, create, and enter records and logical files).
12. Query a database and create reports and/or files from queries.
13. Transfer files between mid-range and microcomputer systems.
14. Implement hardware and software network security solutions (e.g., VPN, SSL, and firewall).
15. Maintain technical industry knowledge.
Competency: Defining System and Business Requirements
Tasks
1. Identify information technology needed to support given sets of tasks and activities for individuals, workgroups, and the organization.
2. Define the role of Information Systems within strategic plan for the total company.
3. Develop a short-range Information System plan and a continuous improvement plan.
4. Determine functional structures (internal vs. outsourcing).
5. Establish goals and objectives for an Information System.
6. Define mission and critical success factors.
7. Formulate Information System operating procedures.
8. Identify hierarchical and flow models of the organization.
9. Define the roles and function of Information System personnel within the organization.
10. Identify drivers and inhibitors of information technology change in the organization.
11. Describe how information technology affects worker-management relationships.
12. Explain how information technology has contributed to worker productivity and teamwork.

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